Exact GNN Architecture

Case star graph + global authority graph for legal outcome prediction

- Current implemented model: hgt with 3 HGT layers, hidden_dim=128, heads=4
- Live graph snapshot: 1010 cases, 181177 total nodes, 360234 forward edges, 720468 directed edges after reverse relations
- Feature construction: 384-d sentence-transformer embedding plus 12 scalar slots per node, total 396 dims
- Purpose of this PDF: explain exactly what every node stores, how the graph is connected, and what happens layer by layer in the current implementation

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Generated from current code and cached graph

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This document is generated from the current implementation and cached graph, not from the old phase-1 note. Source of truth files inspected: configs/gnn_case_star.yaml, data/graph_cache/case_star_global_graph.pt, src/preprocessing/extract.py, src/graph/case_star_builder.py, src/graph/global_graph_builder.py, src/graph/pyg_builder.py, src/models/hetero_gnn.py, and src/training/train.py.

- Generated UTC timestamp: 2026-03-29 08:43:17 UTC
- Model in code/config: architecture=hgt, num_layers=3, hidden_dim=128, num_heads=4, dropout=0.25, mlp_hidden_dim=128
- Graph snapshot in cache: 1010 case nodes, 181177 total nodes, 360234 forward edges, 720468 directed edges after reverse-edge expansion, 18 node types, 50 edge types
- Feature size: 396 = 384 text-embedding dimensions + 12 scalar dimensions
- Split counts on case nodes: train=707, val=151, test=152
- Important correction: the README still says 2 message-passing layers, but the current config and model code use 3 HGT layers.

1. Build Stages

- Stage A - clean each raw case: leakage-bearing fields are removed, decision-like annotations are dropped, and pre-judgment text is assembled into retained sections preamble/facts/arguments.
- Stage B - extract normalized entities: petitioner/respondent/court/judge/lawyer/statute/provision/precedent/org/gpe/date/case_number are read from retained annotations; lawyer mentions are refined into petitioner_lawyer or defence_lawyer when the local text window suggests a side.
- Stage C - build one local case star graph: one case node is created first, then section nodes and entity nodes are attached by typed relations.
- Stage D - merge all local graphs into one global graph: shareable node types are merged by node_key across cases; local-only types keep case-specific keys.
- Stage E - tensorize into PyTorch Geometric HeteroData: every node gets a dense feature vector x, every relation gets an edge_index tensor, and only case nodes receive y/train_mask/val_mask/test_mask.
- Stage F - make the graph bidirectional: ToUndirected() adds a reverse relation for every forward relation so the case node can aggregate back from its neighbors.
- Stage G - run 3 HGT layers and an MLP head: logits are produced only for case nodes, and loss is applied only on training-mask case nodes.

2. Node Types, Scope, and Live Counts

node_type	scope	count	node.text stores
case	case-local	1010	cleaned preamble+facts+arguments concat
preamble	case-local	1010	cleaned preamble text when present
facts	case-local	795	cleaned facts text when present
arguments	case-local	709	cleaned arguments text when present
petitioner	case-local	78007	normalized party string
respondent	case-local	70965	normalized party string
court	shared-global	617	normalized canonical string
judge	shared-global	678	normalized canonical string
petitioner_lawyer	shared-global	939	normalized canonical string
defence_lawyer	shared-global	383	normalized canonical string
lawyer	shared-global	1274	normalized canonical string
statute	shared-global	729	normalized canonical string
provision	shared-global	1700	normalized canonical string
precedent	shared-global	6994	normalized canonical string
org	shared-global	3274	normalized canonical string
gpe	shared-global	4050	normalized canonical string

date	shared-global	4992	normalized canonical string
case_number	shared-global	3051	normalized canonical string

The count column comes from the live cached graph. Preamble exists for every case in the current cache, but facts and arguments are optional in practice and appear only when retained text exists. Petitioner and respondent are intentionally case-local because share_party_nodes=false in the current config.

3. What A Node Stores Before And After Tensorization

- Before PyG tensorization, each node is a GraphNode with fields: node_type, node_key, text, metadata, share_across_cases.
- For section nodes, text is the cleaned section string. For entity nodes, text is the normalized canonical name string. For case nodes, text is the concatenation of retained preamble + facts + arguments.
- During global merge, merged node metadata gains global_case_frequency and then degree.
- After build_pyg_heterodata(), the model-facing HeteroData stores data[node_type].x and data[node_type].node_id for all node types. Only the case store additionally gets y, train_mask, val_mask, test_mask, case_id, file_name, and raw_label.
- The final HeteroData does not carry raw text strings for every node. The raw cleaned texts remain on disk in data/processed/cleaned_cases/*.json; the graph tensors contain numeric features only.

4. Exact Case Node Feature Layout

slot	meaning
0-383	SentenceTransformer embedding of cleaned case text (preamble + facts + arguments)
384	respondent_count
385	judge_count
386	lawyer_count
387	statute_count
388	provision_count
389	precedent_count
390	preamble_length
391	facts_length
392	arguments_length
393	case_year
394	petition_type_known
395	petition_type_hash

The scalar tail on the case node is filled from preprocessing metadata. Concretely this gives the model both a dense semantic summary of the full retained text and a compact numeric summary of selected case-level counts and lengths.

5. Exact Text Node Feature Layout

slot	meaning
0-383	SentenceTransformer embedding of the section text
384	text_length
385	is_preamble
386	is_facts
387	is_arguments
388	cited_statute_count
389	cited_provision_count
390	cited_precedent_count
391	petitioner_lawyer_count
392	defence_lawyer_count

393	petitioner_count
394	respondent_count
395	judge_count

- Observed non-zero scalar columns in the current cache: preamble uses only slots 384 and 385; facts uses only 384 and 386; arguments uses only 384 and 387.
- Slots 388 through 395 exist in the feature vector but are all zero right now because the builder initializes those metadata fields to 0 and never updates them later.
- Because the model is already heterogeneous, the one-hot type slots is_preamble/is_facts/is_arguments duplicate information that is also available from the node type itself.

6. Exact Entity Node Feature Layout

slot	meaning
0-383	SentenceTransformer embedding of the canonical entity string
384	mention_count
385	first_seen_preamble
386	first_seen_facts
387	first_seen_arguments
388	seen_in_arguments
389	seen_in_preamble
390	local_case_frequency
391	global_case_frequency
392	degree
393	is_shared_node
394	padding zero
395	padding zero

- global_case_frequency is written only after the cross-case merge, so it describes how many distinct cases use the merged node.
- degree is computed from the merged graph before PyG conversion and then scaled into the entity scalar tail.
- In the current implementation, slot 384 mention_count and slot 390 local_case_frequency are populated from the same metadata field, so they are exact duplicates.
- Case-local petitioner/respondent nodes keep is_shared_node=0. Shared node types such as court/statute/gpe can still have global_case_frequency=1 if they ended up unique in the current corpus.

7. Exact Edge Storage

- Edges do not have edge_attr in the current pipeline. The graph stores only connectivity and edge type.
- Each relation becomes a separate edge_index tensor of shape [2, E_relation].
- After ToUndirected(), every forward relation gets a reverse relation named rev_<relation>.
- The live cached graph therefore has 25 forward relation types and 50 total directed relation types.

8. Forward Relations And Live Counts

src	relation	dst	count	meaning
case	has_preamble	preamble	1010	case to preamble section
case	has_facts	facts	795	case to facts section, only when facts text exists
case	has_arguments	arguments	709	case to arguments section, only when arguments text exists
case	has_petitioner	petitioner	78007	case to case-local petitioner entity
case	has_respondent	respondent	70965	case to case-local respondent entity
case	heard_in	court	2443	case to normalized court node
case	decided_by_bench	judge	2595	case to normalized judge node

case	has_lawyer	lawyer	8280	case to generic lawyer node
case	has_petitioner_lawyer	petitioner_lawyer	3935	case to petitioner-side lawyer node
case	has_defence_lawyer	defence_lawyer	2753	case to defence-side lawyer node
case	mentions_org	org	19565	case to organization mention node
case	mentions_gpe	gpe	116649	case to location mention node
case	has_date	date	16303	case to date mention node
case	has_case_number	case_number	22816	case to case-number mention node
arguments	cites_statute	statute	1076	arguments section to cited statute
arguments	cites_provision	provision	2396	arguments section to cited provision
arguments	cites_precedent	precedent	3242	arguments section to cited precedent
provision	belongs_to_statute	statute	2219	legal hierarchy link
petitioner_lawyer	citation	arguments	190	petitioner-side lawyer connected to arguments node
defence_lawyer	citation	arguments	66	defence-side lawyer connected to arguments node
provision	used_in_arguments	arguments	2396	shortcut bridge back into arguments
statute	used_in_arguments	arguments	1076	shortcut bridge back into arguments
petitioner	is_party_in_arguments	arguments	441	party bridge into arguments
respondent	is_party_in_arguments	arguments	257	party bridge into arguments
judge	presided_arguments	arguments	50	judge bridge into arguments

There are no dense case-to-case similarity edges in this implementation. Cross-case signal appears only because shared nodes such as court/judge/statute/provision/precedent/org/gpe/date/case_number are reused across cases.

9. Exact Layer-By-Layer Computation

- Input projection step: for every node type t, x_t with shape [N_t, 396] is mapped by a node-type-specific Linear(396 -> 128). Then a learned type embedding of shape [1, 128] is added.
- Layer formula used for each of the 3 message-passing layers: hidden = HGTConv(hidden, edge_index_dict); then for each node type the block does residual + dropout + LayerNorm + ReLU.
- More explicitly, per node type: updated_t = ReLU(LayerNorm(residual_t + Dropout(message_t))).
- Case node at layer 1: it can aggregate only direct neighbors through reverse relations, so it sees preamble/facts/arguments and the attached entity nodes in their layer-0 state.
- Case node at layer 2: this is the first layer where true cross-case information reaches a case node, because shared nodes such as court/statute/provision/judge have already aggregated from all attached cases during layer 1.
- Case node at layer 3: it receives neighbor states that already contain two rounds of contextual mixing, so deeper patterns like case <- shared authority <- other case <- that case's local neighborhood can influence the representation.
- Examples of meaningful layer-2 and layer-3 routes: case_A <- court <- case_B, case_A <- arguments_A <- statute <- arguments_B, case_A <- judge <- case_B, case_A <- arguments_A <- provision <- arguments_B.

10. HGT Block And Classification Head

- HGT block configuration: HGTConv(in_channels=128, out_channels=128, heads=4).
- The model is heterogeneous at every layer: relation type and node type both matter, so each relation gets relation-specific attention projections inside HGTConv.
- Classifier head on case nodes only: Linear(128 -> 128) -> ReLU -> Dropout(0.25) -> Linear(128 -> 2).
- Output logits are produced for every case node, but loss is computed only on case nodes where train_mask is True.

11. Training Procedure

- Device is cuda if available, otherwise cpu.
- Optimizer: AdamW with lr=0.001 and weight_decay=1e-05.
- Balanced class weights are computed from only the training case labels when class_weight=balanced.
- At each epoch, the model runs on the full graph; only the supervision mask changes which case nodes contribute to loss and which contribute to validation/test metrics.

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- Early stopping is enabled with patience=15 over validation macro F1.
- Training budget in config: epochs=60, log_every=5.
- Best checkpoint selection uses validation macro F1; the saved best state is reloaded before final
  train/val/test evaluation.

12. Important Implementation Notes
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- Facts and arguments nodes are optional, not guaranteed. In the current cache there are 795 facts nodes
  and 709 arguments nodes for 1010 cases.
- If a provision mention references a statute but the statute node is otherwise absent, the builder can
  create a synthetic statute node so belongs_to_statute still has a target.
- The current SentenceTransformer backend ignores the max_length value from the config in this code
  path; truncation behavior is governed by the encoder implementation itself, not by an explicit
  max_length argument here.
- Approximate live tensor footprint from node features alone is about 273.69 MB; edge_index tensors add
  about 10.99 MB more.
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